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RESEARCH ARTICLE

LAC: A Lightweight Linear-Attention Framework for Efficient Learned Image Compression

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ABSTRACT Efficient learned image compression is critical for modern imaging applications, which demand a balance between high fidelity, low latency, and computational cost, especially on resource-constrained devices. To address these challenges, we introduce LAC, a Lightweight Attention-based Compression framework. Our contributions are threefold: 1) a novel Parallel Attention–Convolution (PAC) module that efficiently captures both local and global features using a lightweight convolutional branch and a linear-attention branch; 2) a Channel-Gated (CG) unit that reduces entropy coding redundancy; and 3) a Shared Head Matrix (SHM) mechanism that decreases parameters in the attention heads. Extensive experiments show that LAC achieves rate-distortion performance competitive with state-of-the-art (SOTA) methods while being significantly more efficient, reducing the parameter count by 40% and improving inference speed by 13.8% compared to a leading SOTA model. By deliberately prioritizing computational efficiency and low latency over peak rate-distortion scores, LAC presents a practical and effective solution for high-quality, efficient image compression, making it a practical solution for low-latency media and mobile applications where computational resources are limited.

INDEX TERMS Learned image compression, lightweight transformer, linear attention, parallel attention, efficient image compression.

I. INTRODUCTION

The proliferation of high-resolution imaging on mobile and edge devices has created an urgent need for efficient image compression. Many applications, from social media content delivery to remote collaboration, operate under strict constraints, including low uplink bandwidth and demanding latency requirements. Effective image compression is therefore crucial, not only for reducing transmission and storage costs but also for preserving the integrity of critical visual details. However, existing solutions fall short: traditional codecs like JPEG [1] and JPEG2000 [2] often introduce blocking artifacts that obscure details, while more advanced standards like VVC Intra [3] can be too computationally demanding for low-power mobile devices. While deep learning-based methods have shown great promise, develop-

ing a solution that is both lightweight and powerful enough for a wide range of resource-constrained environments remains an open challenge.

A significant research gap exists in developing a compression framework that is both powerful and efficient enough for this demanding scenario. On one hand, methods based on Convolutional Neural Networks (CNNs) [4] are computationally efficient but struggle to model the long-range dependencies required to understand the global context of complex visual scenes. On the other hand, Transformer-based approaches [5], [6], which have revolutionized many fields, suffer from quadratic computational complexity and large parameter counts when applied to vision tasks. This makes them impractical for latency-sensitive processing on resource-constrained mobile devices. This trade-off between local efficiency and global modeling capability highlights the need for a novel architecture that can bridge this gap. Unlike research focused on maximizing rate-distortion performance,

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our work is explicitly aimed at developing a lightweight framework optimized for the low-latency and low-compute environments typical of mobile and edge devices.

To address these challenges, this paper proposes the **Lightweight Attention-based Compression (LAC)** framework. While our work follows the established CNN-Transformer hybrid paradigm, its novelty lies in improving efficiency and providing a new performance-complexity trade-off, rather than redefining the coding pipeline. The framework is specifically designed to balance rate-distortion performance, efficiency, and applicability in resource-constrained environments, and our key contributions are as follows:

- We introduce a *Parallel Attention–Convolution (PAC)* module, a hybrid block that integrates sliding-window linear attention with partial and depthwise separable convolutions. This design efficiently captures both local and global features while significantly reducing computational overhead.
- We propose two novel optimization units: a *Channel-Gated (CG)* unit to mitigate inter-channel redundancy in the entropy model, and a *Shared Head Matrix (SHM)* mechanism to reduce redundant computations in the multi-head attention blocks.
- We demonstrate through extensive experiments that LAC achieves rate-distortion performance competitive with state-of-the-art methods, while reducing the parameter count by approximately 40% and improving inference speed by 13.8%, making it a practical solution for efficient media delivery on resource-constrained devices.

II. RELATED WORK

Deep learning has played a pivotal role in advancing image compression, offering improved R-D performance and adaptability compared to traditional codecs. This section reviews the main paradigms in learned image compression, discusses the evolution of attention mechanisms, and analyzes the specific challenges in achieving efficient, high-fidelity compression that motivate this work.

A. DEEP LEARNING-BASED IMAGE COMPRESSION PARADIGMS

Three main paradigms, based on Variational Autoencoders (VAEs), CNNs, and Vision Transformers (ViTs), have shaped the landscape of learned image compression. VAE-based methods [7], [8] excel at creating continuous latent spaces, enabling robust reconstruction suitable for media content. CNNs remain a foundational architecture due to their strong local feature extraction capabilities, with models like [4] and [9] outperforming traditional codecs in perceptual quality by efficiently preserving localized details. More recently, ViTs [10], [11] have emerged as a competitive alternative, using attention mechanisms to model non-local dependencies. Hybrid approaches, such as LIC_TCM [6],

combine CNNs and Transformers to leverage both local and global modeling strengths, achieving state-of-the-art R-D performance on multiple benchmarks. This evolution reveals a clear trend: combining the local efficiency of CNNs with the global modeling power of Transformers is a necessary step for SOTA performance, which justifies our decision to develop a novel hybrid architecture. Beyond these, the field is also expanding into novel paradigms. For instance, some work focuses on unified compression frameworks for both human and machine vision [12], while other approaches leverage implicit neural representations to achieve high compression ratios [13]. However, the former addresses a different problem scope, and the latter often entails significant decoding complexity, rendering them less suitable for the strict low-latency demands of efficient media delivery on edge devices.

B. ATTENTION MECHANISMS AND EFFICIENCY OPTIMIZATIONS

The success of Transformers is largely due to the self-attention mechanism [5], which excels at modeling long-range dependencies. However, its computational complexity scales quadratically ($O(N^2)$) with the number of input tokens N , creating a significant bottleneck for high-resolution images. This has spurred research into more efficient attention variants. A prominent solution is Linear Attention [14], which reduces complexity to linear time ($O(N)$) by approximating the softmax function with kernel methods. This optimization is critical for accelerating inference and enabling the processing of large-scale inputs, making it highly suitable for real-time applications. However, directly replacing softmax attention with a simple linear variant can lead to a drop in performance [15]. This trade-off between computational efficiency and expressive power is a central challenge that existing works have yet to fully resolve. Our work confronts this issue directly by proposing the PAC module, which is engineered to balance these competing demands.

C. CHALLENGES IN EFFICIENT LEARNED IMAGE COMPRESSION

While general-purpose compression methods have achieved impressive R-D gains, deploying them on resource-constrained devices imposes unique constraints that are often unaddressed. The primary requirements are ultra-low latency, a minimal computational load, and the preservation of fine-grained details like text and textures, which can be easily degraded by aggressive compression. For instance, the limited receptive fields of purely CNN-based methods [4] can fail to capture the global context of a complex scene, potentially sacrificing the integrity of important non-local relationships. Conversely, state-of-the-art models like LIC_TCM [6], while achieving excellent R-D scores, rely on complex architectures with quadratic-cost attention mechanisms, making them unsuitable for many real-time,

TABLE 1. Comparison of different compression approaches.

Method	Core Technique	Complexity	Strengths & Limitations	Suited for Mobile Use?
BPG [16]	HEVC Intra-frame	N/A	High quality; Static, non-adaptive.	No (Non-adaptive)
Cheng [4]	CNN + Attention	N/A	Good perceptual quality; Limited global context.	Partially (Lacks global context)
LIC_TCM [6]	Transformer + CNN	Quadratic	SOTA R-D performance; High complexity.	No (High complexity)
Ours (LAC)	PConv + Linear Attn.	Linear	Fast, low-params; Optimized for mobile efficiency.	Yes

resource-constrained environments. This highlights a critical gap: the need for a compression framework specifically engineered for high-speed, high-fidelity, and resource-sensitive applications. Our work directly addresses this gap by designing a framework that prioritizes computational efficiency and parameter reduction without significantly compromising on visual quality. A summary comparing these different approaches is presented in Table 1.

III. METHOD

A. FRAMEWORK ARCHITECTURE

The proposed LAC framework is built upon the established VAE paradigm [17], which is optimized end-to-end for a target R-D trade-off. As illustrated in Figure 1, the architecture employs a main encoder-decoder pair (g_a, g_s) and a hyperprior encoder-decoder pair (h_a, h_s).

The overall process is as follows:

- 1) The main encoder g_a transforms an input image x into a latent representation y .
- 2) The hyperprior encoder h_a further compresses y into a hyper-latent representation z to capture spatial redundancies in y .
- 3) Both representations are quantized ($\hat{y}, \hat{z}$) and entropy-coded. The hyper-latent $\hat{z}$ is used to parameterize a conditional entropy model [6], [17] for $\hat{y}$, minimizing the bitrate.
- 4) Finally, the main decoder g_s reconstructs the image $\hat{x}$ from the quantized latent $\hat{y}$.

The framework is trained by minimizing the R-D loss function:

$$\begin{aligned}
 J = & \mathbb{E}_{X \sim p_X} [-\log_2 p_{\hat{y}}(\hat{y} | \hat{z})] \\
 & + \mathbb{E}_{X \sim p_X} [-\log_2 p_{\hat{z}}(\hat{z})] \\
 & + \lambda \cdot \mathbb{E}_{X \sim p_X} [d(x, \hat{x})]
 \end{aligned} \quad (1)$$

where the first two terms represent the bitrate (R) and the third term measures the distortion (D) between the original image x and the reconstructed image $\hat{x}$. The hyperparameter λ controls the R-D trade-off; a set of models is trained with different λ values to generate a full performance curve, with smaller values prioritizing quality (low distortion) and larger values prioritizing file size (low rate).

While this architecture follows a standard paradigm, our primary contributions lie in the novel designs of the encoder-decoder networks and the optimization modules. Specifically, we introduce: 1) the PAC module for efficient feature extraction, 2) the CG unit for enhancing the entropy

model, and 3) the SHM mechanism for optimizing attention heads. These components are detailed in the subsequent sections.

B. A HYBRID APPROACH FOR FEATURE EXTRACTION VIA PAC MODULE

In deep learning-based image compression, particularly for applications on resource-constrained devices, a model must efficiently handle two distinct types of visual information. Natural images often contain a composite of crucial *local features*—such as the text on a sign, a person’s facial expression, or logos—where any loss of detail can compromise visual fidelity. Simultaneously, they require preserving the *global context*, like the overall scene of an event, to maintain narrative integrity. This dual requirement motivates our design of the PAC module, a hybrid block that serves as the core component of our encoders and decoders.

As depicted in Figure 2, the PAC module is engineered to efficiently model both feature types. It first employs a *convolutional embedding* stage to tokenize the input image x into a compact latent representation. By embedding spatial resampling within strided convolutions, this approach offers two key advantages for our target application. First, since convolution implicitly encodes positional cues, explicit positional embeddings [18] become unnecessary, reducing model complexity for mobile deployment. Second, unlike methods based on non-overlapping pixel blocks [19], our convolution-based tokenization mitigates the block artifacts that can severely degrade the readability of text in images. Following this embedding, the features are fed into parallel branches for local and global feature extraction, which are detailed in the subsequent subsections.

1) PARALLEL ARCHITECTURE FOR LOCAL-GLOBAL FUSION

To efficiently model both local and global features, the PAC module is built upon a series of carefully integrated techniques. The foundation is an efficient attention mechanism, which is progressively enhanced to balance performance and computational cost.

a: LINEAR ATTENTION MECHANISM

Standard self-attention suffers from quadratic complexity ($O(N^2)$), creating a bottleneck for high-resolution image compression. To overcome this, we adopt a linear attention mechanism [14], which reduces complexity to $O(N)$. As shown in Equations 2-4, this is achieved by approximating the similarity matrix with a kernel mapping $\phi(\cdot)$ and

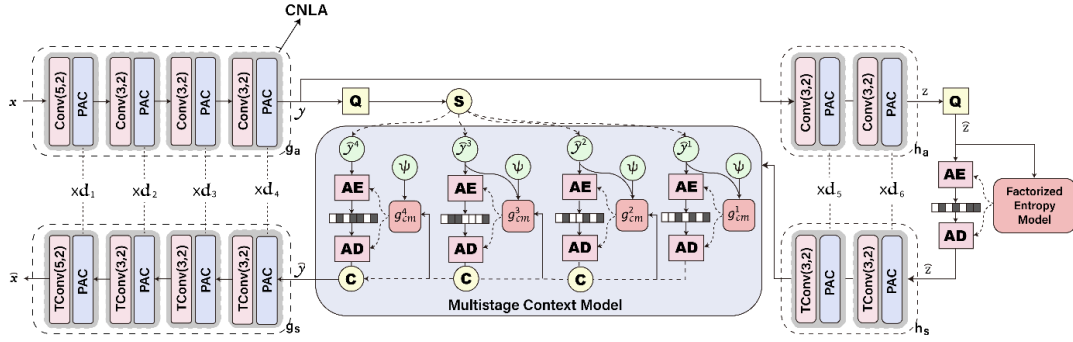


FIGURE 1. Overview of the proposed LAC framework.

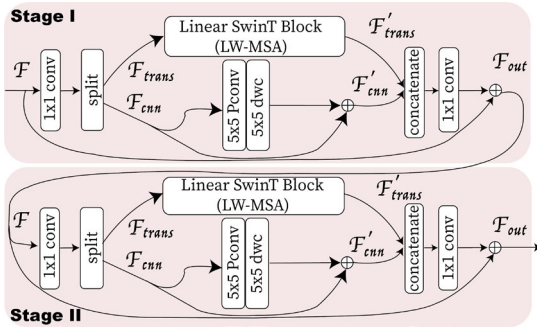


FIGURE 2. The architecture of the proposed PAC module.

leveraging the associativity of matrix multiplication. This removes the explicit $N \times N$ similarity matrix, significantly cutting time and memory costs (see Figure 3). This efficiency gain is the first critical step toward making a Transformer-based model viable for efficient compression on mobile devices.

$$\text{Sim}(Q, K) = \phi(Q)\phi(K)^T \quad (2)$$

$$O_i = \sum_{j=1}^N \frac{\phi(Q_i)\phi(K_j)^T}{\sum_{j=1}^N \phi(Q_i)\phi(K_k)^T} V_j \quad (3)$$

$$O_i = \frac{\phi(Q_i)(\sum_{j=1}^N \phi(K_j)^T V_j)}{\phi(Q_i)(\sum_{j=1}^N \phi(K_j)^T)} \quad (4)$$

b: ENHANCING LINEAR ATTENTION

Although computationally efficient, directly replacing standard attention with a basic linear variant can lead to performance degradation [15], [20], [21]. To address this, we enhance its expressiveness by incorporating two techniques inspired by Han et al. [22]. First, we introduce a mapping function $\phi_p(x)$ to refine the similarity computation, as shown in Equations 5-6 and Figure 4. Second, we serially add a DWC as a rank restoration module (Figure 5). As formulated in Equations 7-8, DWC acts as a localized attention mechanism, increasing the rank bound of the equivalent attention map M_{eq} . In practical terms, this ensures that while the model remains computationally efficient,

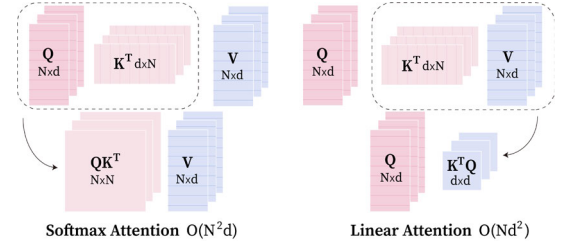


FIGURE 3. Comparison between traditional softmax attention and linear attention in terms of computation structure.

it does not lose the ability to distinguish and preserve fine-grained details, such as the texture of a fabric or the subtle features of a face in a photograph.

$$\text{Sim}(Q_i, K_j) = \phi_p(Q_i)\phi_p(K_j)^T \quad (5)$$

$$\phi_p(x) = f_p(\text{ReLU}(x)), \quad f_p(x) = \frac{\|x\|}{\|x^{**p}\| \cdot x^{**p}} \quad (6)$$

Here, x^{**p} denotes the element-wise power operation, following the convention introduced in [22]. As shown in Figure 4, ReLU ensures non-negativity for numerical stability in Eq. 6.

$$O = \phi(Q)\phi(K)^T V + \text{DWC}(V) \quad (7)$$

$$O = (\phi(Q)\phi(K)^T + M_{\text{DWC}})V = M_{\text{eq}}V \quad (8)$$

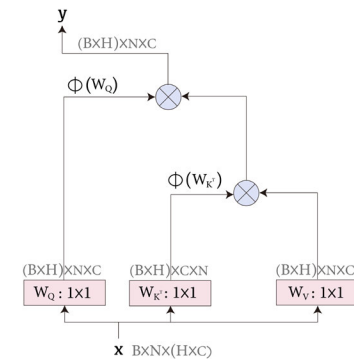


FIGURE 4. Computation flow of linear attention using mapping functions.

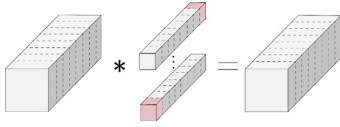


FIGURE 5. Computation diagram of DWC.

c: PARALLEL BLOCK FOR EFFICIENCY

While the serial DWC enhancement boosts performance, it introduces a new computational bottleneck [23], with frequent memory accesses that can slow down training by over 3x (see Table 2). To resolve this, we adopt a parallel structure inspired by [6], splitting the input channels to be fed into a convolutional branch and a linear attention branch simultaneously (Figure 6). This parallel design, summarized in Equation 9, allows the module to concurrently analyze an image for broad, scene-level context (via the attention branch) and fine, localized details (via the convolution branch), ensuring that neither type of information is lost while mitigating the training slowdown.

$$\begin{aligned} F_{DWC}, F_{Atten} &= \text{Split}(\text{Conv}_{1 \times 1}(F)), \\ F'_{DWC}, F'_{Atten} &= \text{DWC}(F_{DWC}), \text{LinearA}(F_{Atten}), \\ F_{out} &= F + \text{Conv}_{1 \times 1}(\text{Cat}(F'_{DWC}, F'_{Atten})). \end{aligned} \quad (9)$$

TABLE 2. Training time comparison with and without DWC.

Method	Training Time (s/epoch)
w DWC	612
w/o DWC	193

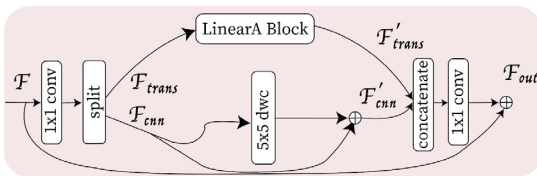


FIGURE 6. PLD module structure.

d: PCONV FOR ENHANCED LOCAL REPRESENTATION

Finally, to further improve the local feature modeling within the parallel block, we integrate PConv [23] as a preprocessing step before the DWC operation. PConv applies convolution to only a subset of channels, strengthening inter-channel information modeling while reducing computational overhead. This targeted convolutional processing is particularly effective at preserving the sharpness of edges and the clarity of text, which are often the first details to degrade in standard compression, thus directly improving the perceptual quality of the transmitted image.

2) SLIDING WINDOW LINEAR ATTENTION FOR EFFICIENT GLOBAL MODELING

While the PAC module's parallel design is efficient, applying linear attention to a full high-resolution feature map can still be computationally demanding. To address this, we introduce a key innovation: combining the sliding window mechanism, inspired by the Swin Transformer [24], with our *linear* attention model. Standard Swin Transformer applies computationally intensive quadratic attention within each local window. Our approach, in contrast, uses the highly efficient linear attention, creating a doubly efficient model. This synergistic combination significantly reduces the computational cost, making the powerful window-based attention paradigm practical for the first time in a lightweight compression framework suitable for mobile deployment. This strategy restricts self-attention to smaller, non-overlapping local windows, which drastically lowers the sequence length processed by the attention layer without compromising the model's ability to learn global context.

a: WINDOWED AND SHIFTED-WINDOW ATTENTION

The mechanism operates by alternating between two types of attention blocks across consecutive layers, as illustrated in Figure 7. In the first block, standard Local Window Multi-head Self-Attention (LW-MSA) is applied. The feature map is partitioned into uniform, non-overlapping windows (e.g., 4×4 pixels), and self-attention is computed only within each window. This step efficiently models local information.

To enable information flow between these windows and build a global context, the subsequent block employs Shifted Local Window Multi-head Self-Attention (SLW-MSA). In this block, the window partitioning is shifted by half a window size (e.g., $\lfloor M/2 \rfloor$ pixels) relative to the previous layer. This new configuration creates connections across the boundaries of the previous layer's windows, allowing for cross-window information exchange. The formal transformation across two such consecutive blocks is defined as:

$$\begin{aligned} \hat{z}^l &= \text{LW-MSA}(\text{LN}(z^{l-1})) + z^{l-1}, \\ z^l &= \text{MLP}(\text{LN}(\hat{z}^l)) + \hat{z}^l, \\ \hat{z}^{l+1} &= \text{SLW-MSA}(\text{LN}(z^l)) + z^l, \\ z^{l+1} &= \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1}. \end{aligned} \quad (10)$$

Here, $\hat{z}^l$ and z^l are the outputs from the (S)LW-MSA and MLP modules at the l -th block.

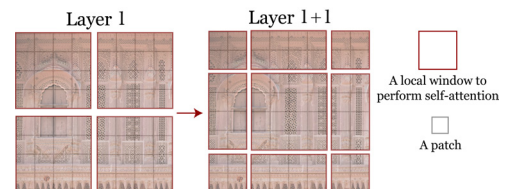


FIGURE 7. Workflow of the sliding window mechanism.

b: EFFICIENT IMPLEMENTATION WITH CYCLIC SHIFT

A naive implementation of shifted windows would generate windows of varying sizes and increase the total number of windows, making batch computation inefficient. To solve this, an efficient batching scheme based on a *cyclic shift* is adopted, as shown in Figure 8. By cyclically shifting the feature map upwards and to the left, the shifted windows become aligned in memory. However, this causes some batched windows to contain non-adjacent sub-windows from the original feature map. To ensure that attention is only computed within these valid sub-windows, a *masking mechanism* is applied to filter out attention between disconnected regions. This approach preserves computational efficiency while enabling the powerful cross-region feature interaction of the shifted window design.

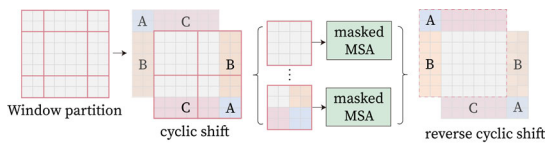


FIGURE 8. Efficient batched computation for shifted window partitioning in self-attention.

C. ENTROPY CODING OPTIMIZATION VIA CG UNIT

In modern learned image compression, high performance is achieved via adaptive entropy models that condition the probability of latent symbols on both hyperpriors and previously decoded latents [11], [17], [25]. A key technique for this is exploiting inter-channel redundancies, often through channel grouping strategies [4], [26], which range from uniform grouping [27], [28] to channel-by-channel approaches [29], [30]. A particularly effective strategy is non-uniform channel conditional coding [31], which assumes that earlier channels are generally more informative and assigns coding resources accordingly (Figure 9).

However, this reliance on a fixed, predefined channel importance order is a fundamental limitation. As validated in prior work [27] and illustrated in our case study (Figure 10), this assumption is often violated; some later-encoded channels can exhibit high entropy, while some earlier ones may be redundant. This mismatch between the fixed model assumption and the actual content leads to suboptimal bit allocation and wasted coding efficiency.

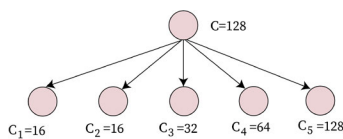


FIGURE 9. Illustration of the non-uniform grouping method for adaptive coding of channel conditions.

To overcome this limitation, we propose the CG unit, a lightweight and trainable module that introduces a dynamic, content-aware channel scaling mechanism. Instead of relying

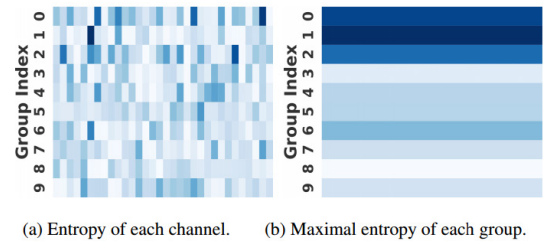


FIGURE 10. Case study of adaptive coding for 10-slice channel conditions. Darker colors indicate higher entropy values.

on a fixed grouping scheme, the CG unit adaptively reweights each channel based on its learned importance before the context modeling stage. As depicted in Figure 11, the CG unit consists of a small set of learnable parameters, W , initialized to one, which scale the amplitude of each channel in the quantized latent representation $\hat{y}$:

$$\hat{y}_i = W_i \hat{y}_i \quad (11)$$

where i is the channel index and W_i is the learned weight for that channel. This simple yet powerful mechanism allows the model to allocate bits more intelligently, dynamically emphasizing informative channels and suppressing redundant ones based on the specific image content. As demonstrated in our ablation studies (Section IV-C), this results in a measurable improvement in R-D performance with virtually no impact on decoding speed, highlighting its practical value for efficient, high-fidelity compression.

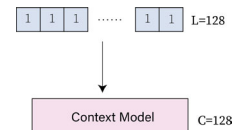


FIGURE 11. Structure of the proposed CG unit.

D. ATTENTION HEAD OPTIMIZATION VIA SHM MECHANISM

The Multi-Head Self-Attention (MHSA) mechanism, a cornerstone of the Transformer architecture [5], [32], enhances model performance by capturing diverse feature dependencies in parallel (Figure 12). However, this strength comes at a cost: MHSA often leads to significant redundancy, where multiple attention heads learn to attend to similar or even identical patterns. In the context of lightweight compression for mobile devices, this redundancy is particularly detrimental, as it inflates the parameter count and wastes computational cycles without contributing to performance.

To address this head-wise redundancy, we propose the SHM mechanism. The core idea is to decompose the projection matrices within each attention head into two components: a single matrix H that is shared across all heads, and a smaller, head-specific residual matrix. As depicted in Figure 13, the shared matrix learns foundational patterns

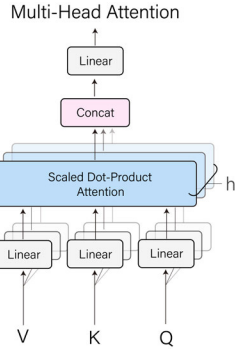


FIGURE 12. Multi-head attention mechanism.

common to all heads, while each head only needs to learn a lightweight residual. This approach offers two significant advantages over standard MHSA. First, by factoring out common patterns, it drastically reduces the number of unique parameters, mitigating redundancy and improving parameter efficiency. Second, unlike cascading methods like Cascade Group Attention (CGA) [33] that introduce serial dependencies, SHM fully preserves the parallel computation of MHSA, ensuring that the optimization does not introduce additional latency.

Consequently, the SHM mechanism leads to a more efficient and faster attention block, making it highly suitable for efficient media and edge-device delivery. The effectiveness of this approach in reducing redundancy is empirically validated in our experiments (see Section IV-C2), where cross-head similarity heatmaps show that the attention patterns produced by SHM are significantly more diverse than those from standard MHSA.

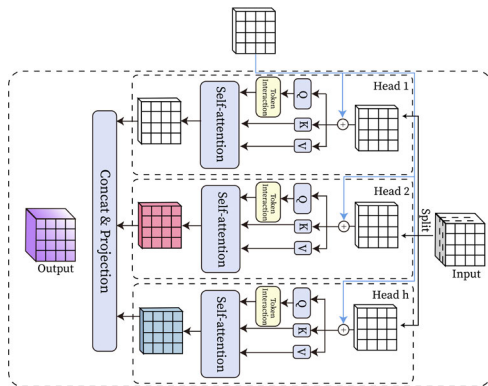


FIGURE 13. Illustration of the proposed SHM mechanism.

E. SUMMARY AND COMPARISON TO SOTA METHODS

To contextualize our contributions, it is crucial to distinguish LAC from related state-of-the-art frameworks. While our work builds upon the efficiency of linear attention, our primary architectural innovation, the PAC module, differentiates itself from standard **Linear Transformers** by integrating a parallel convolutional branch. This hybrid

design allows LAC to simultaneously capture local texture and global context, a feature not present in typical models that rely solely on the attention mechanism. Furthermore, our optimization strategy diverges significantly from other hybrid models like **LIC_TCM**. Whereas **LIC_TCM** employs a computationally heavier cross-attention mechanism for entropy modeling, our framework introduces two targeted, lightweight optimizations: the CG unit, which prunes inter-channel redundancy, and the SHM mechanism, which reduces parameter duplication within the attention heads themselves. This focus on lightweight, modular optimizations is central to our goal of achieving high efficiency on constrained devices.

TABLE 3. Comparison of key components with SOTA methods.

Component	Our Method (LAC)	Compared Method
Global & Local Feature Modeling	PAC Module (Parallel Linear Attention + PConv)	Swin-Transformer (Windowed Quadratic Attn.)
Entropy Model Optimization	CG Unit (Adaptive channel scaling)	LIC_TCM (Cross-Attention Model)
Attention Head Optimization	SHM (Shared Head Matrix)	Standard MHSA (Independent Parameters)

F. COMPLEXITY ANALYSIS

The primary motivation for our architectural choices is to reduce the computational complexity of the attention mechanism, which is a major bottleneck in Transformer-based compression models. Standard self-attention, when applied to a latent representation of size $H \times W$ with d channels, has a computational and memory complexity of $O((HW)^2d)$ due to the quadratic scaling with the number of tokens $N = HW$. Our LAC framework, by contrast, fundamentally reduces this. The core PAC module employs linear attention, which lowers the complexity to be linear with respect to the number of tokens, resulting in $O(HWd^2)$. Furthermore, the sliding window mechanism within the PAC module computes attention only within local windows of size $M \times M$, further reducing the effective number of tokens in each computation. The CG and SHM modules add a negligible number of parameters and operations. This theoretical reduction from quadratic to linear complexity is the primary driver of the efficiency gains reported in our experiments, enabling the model to be practical for resource-constrained environments without requiring hardware-specific optimizations.

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

1) DATASETS

We use the large-scale Flickr2W dataset [33] as the training corpus, which contains 20,000 high-quality images. For testing, we evaluate our model on two standard benchmarks: the *Kodak* dataset, which consists of 24 uncompressed images of size 768×512 , and the *CLIC Professional Validation* dataset [34], which contains 412 high-resolution

images. Kodak and CLIC are widely used as benchmarks and contain significant content—such as faces, street scenes, and text signage—that are common in many real-world imaging applications. This makes them suitable for evaluating the generalization performance of our framework.

2) TRAINING DETAILS

During training, all images are randomly cropped into $256 \times 256 \times 3$ patches. We allocate 99% of the samples for training and the rest for validation. The models are trained for 400 epochs using the Adam optimizer [35] with a batch size of 8 on a single NVIDIA RTX 3090 GPU. The initial learning rate is set to 1×10^{-4} and decays to 1×10^{-5} after 300 epochs. Following the standard end-to-end optimization objective in Eq. 1, we train separate models for two distortion metrics: Mean Squared Error (MSE) [36] and MS-SSIM [37]. The trade-off parameter λ is set to {0.0018, 0.0035, 0.0067, 0.013, 0.025, 0.0483} for MSE-optimized models, and {3, 5, 8, 16, 36} for MS-SSIM-optimized models.

3) EVALUATION PROTOCOL

We measure compression performance using PSNR and MS-SSIM as the primary objective metrics. For the Bjontegaard-Delta rate (BD-rate) [38] evaluation, PSNR is computed on the RGB channels (PSNR-RGB). Crucially, to evaluate the model's suitability for real-world deployment, we also place significant emphasis on efficiency metrics. These include model complexity (parameter count) and inference speed (encoding/decoding time), which are critical indicators of feasibility on resource-constrained mobile devices and in the low-latency pipelines required for media content transmission. For fair comparison with other methods, all inference speed tests were conducted on an NVIDIA RTX 3090 GPU. However, the model's lightweight design is explicitly intended to enable deployment on mobile SoCs and edge GPUs, the performance on which will be explored in future work.

B. R-D PERFORMANCE COMPARISON

We benchmark our proposed method against classical codecs (BPG, VVC VTM 12.1), SOTA learned methods (Balle [11], Cheng [4], Chen [39], He [31], LIC_TCM [6]), and a baseline model using standard sliding window attention [24]. Among these, we select LIC_TCM as the primary baseline for a detailed analysis. This is because it represents the most relevant state-of-the-art method that also adopts a CNN-Transformer hybrid architecture, sharing the same paradigm of attention-based latent transform and entropy modeling as our work. Therefore, using LIC_TCM as the main point of comparison enables a fair and architecture-consistent evaluation that best isolates the benefits of our proposed PAC, CG, and SHM modules. While other methods are included in the rate-distortion curves for

broader context, the core architectural comparison is most meaningful against LIC_TCM.

As shown by the R-D curves in Figure 14, our method demonstrates consistently strong performance across both the Kodak and CLIC datasets. It significantly outperforms conventional codecs like BPG and VTM at all bitrates. When compared to other learned methods, our approach achieves highly competitive PSNR and MS-SSIM scores. Notably, in the low-bitrate region—which is most critical for bandwidth-constrained applications—our model's performance matches and even surpasses the current SOTA method, LIC_TCM. While LIC_TCM shows a slight advantage at high bitrates, our model's superior performance at low bitrates underscores its suitability for practical, real-world applications.

The primary advantage of our model, however, lies in its exceptional efficiency. As detailed in Table 4, our model requires only **349M** parameters, a **40% reduction** compared to LIC_TCM's 877M. Furthermore, it achieves a **13.8% improvement in inference speed**. This significant reduction in both model complexity and latency, combined with its strong low-bitrate performance, highlights our framework's value for efficient media and edge-device delivery.

TABLE 4. Complexity and inference speed comparison with LIC_TCM.

Method	Parameters (M)	Enc (s)	Dec (s)
Ours	349	0.123	0.131
LIC_TCM	877	0.151	0.152

To provide a final, holistic benchmark, we report the BD-rate savings relative to the BPG codec in Table 5. Our method achieves significant BD-rate reductions of **-28.33%** on Kodak and **-28.12%** on CLIC. These results are highly competitive, approaching the SOTA performance of LIC_TCM (-30.54% and -29.12%) while, as previously noted, operating with substantially lower complexity. This confirms that our approach provides a superior trade-off between compression efficiency and computational cost.

TABLE 5. BD-rate on different datasets.

Method	Kodak	CLIC
Balle (ICLR'18)	5.40	8.96
Cheng (CVPR'20)	-17.15	-12.61
Chen (AAAI'22)	-25.11	-22.89
He (CVPR'22)	-25.92	-20.76
LIC_TCM (CVPR'23)	-30.54	-29.12
VTM	-19.71	-18.80
Ours	-28.33	-28.12

C. ABLATION STUDIES

In this section, we conduct ablation studies to isolate and evaluate the contribution of our proposed modules. We systematically analyze the impact of each component in the PAC module, as well as the effectiveness of the CG and SHM units, on both rate-distortion performance and computational complexity.

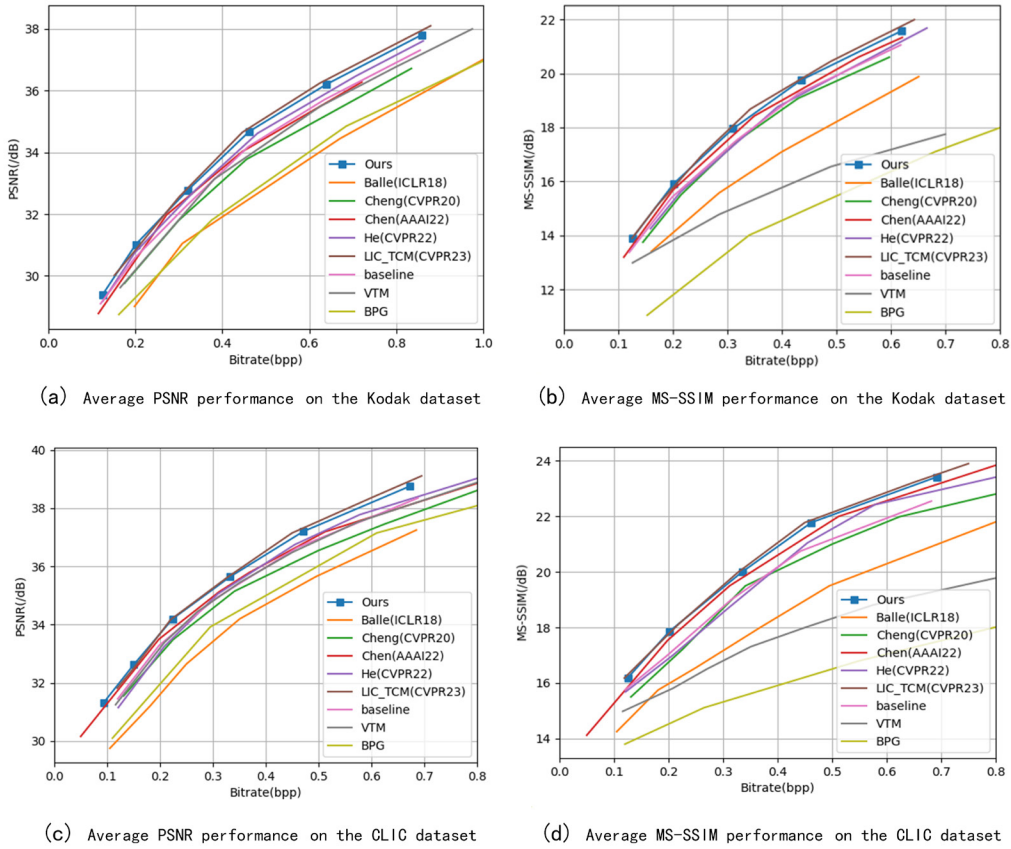


FIGURE 14. Average PSNR/MS-SSIM performance of different compression methods on the Kodak/CLIC dataset.

1) VALIDATION OF PAC MODULE COMPONENTS

To assess the contribution of each component within the PAC module, we adopt a progressive build-up strategy, with results presented in Table 6. The analysis reveals a clear narrative of balancing performance and efficiency. The initial move from a standard attention baseline to our **Linear** model confirms the core trade-off: while PSNR drops slightly (0.21 dB), inference speed more than doubles, with the encoding latency being particularly critical as it directly impacts the upload speed in bandwidth-constrained mobile environments, establishing linear attention as a critical foundation for efficiency. Building on this, the **DLinear** and **DPLinear** configurations show that introducing the parallel architecture and the PConv module, respectively, yields progressive PSNR gains (0.12 dB and 0.29 dB) with only marginal increases in latency. This validates their role in efficiently enhancing feature extraction. Finally, the transition from **DPLinear** to **DPLinearSwinT** (our full PAC module) provides the final performance boost. It achieves the highest PSNR with a negligible increase in inference time, confirming that the sliding window mechanism is an effective optimization that improves performance without sacrificing the efficiency gains of the underlying linear attention.

TABLE 6. Impact of PAC components on compression performance.

Method	bpp	PSNR	Enc(s)	Dec(s)
baseline	0.316	32.22	0.201	0.205
Linear	0.315	32.01	0.092	0.091
DLinear	0.317	32.13	0.101	0.098
DPLinear	0.318	32.42	0.105	0.104
DPLinearSwinT	0.319	32.45	0.106	0.105

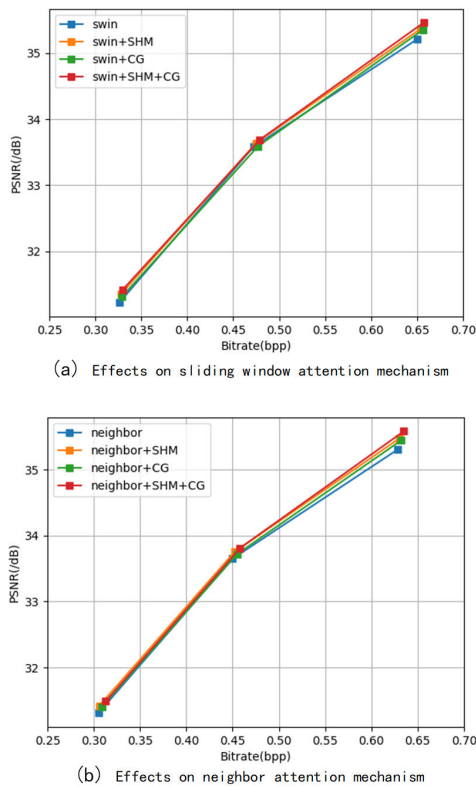
2) VALIDATION OF THE EFFECTIVENESS OF CG AND SHM

We next isolate the contributions of the CG and SHM optimization units by applying them to the finalized PAC model. The results on the Kodak dataset, summarized in Table 7, clearly demonstrate their independent and combined value. Incorporating the *CG unit* alone improves PSNR by 0.10 dB by enabling more efficient bit allocation, while adding the *SHM mechanism* alone improves it by an even greater 0.19 dB by reducing head-wise redundancy. When used together, their effects are complementary, achieving the best overall performance (32.77 dB). Crucially, these significant gains in reconstruction quality are achieved with no measurable impact on encoding or decoding times, highlighting that CG and SHM are highly effective, lightweight optimizations.

TABLE 7. Impact of SHM and CG on compression performance.

Method	bpp	PSNR	Enc(s)	Dec(s)
PAC	0.316	32.45	0.106	0.105
PAC+CG	0.317	32.55	0.106	0.106
PAC+SHM	0.318	32.64	0.107	0.106
PAC+CG+SHM	0.320	32.77	0.107	0.106

To further underscore their value, we test their generalization capabilities by applying them to models built with different attention mechanisms. As shown in Figure 15, CG and SHM consistently enhance reconstruction quality across all tested variants, with their combined use yielding the most significant gains. This demonstrates that they are robust and broadly applicable modules for improving the efficiency of attention-based compression frameworks.

**FIGURE 15. Effectiveness of the CG and SHM modules across different attention mechanisms.**

D. VISUAL QUALITY COMPARISON

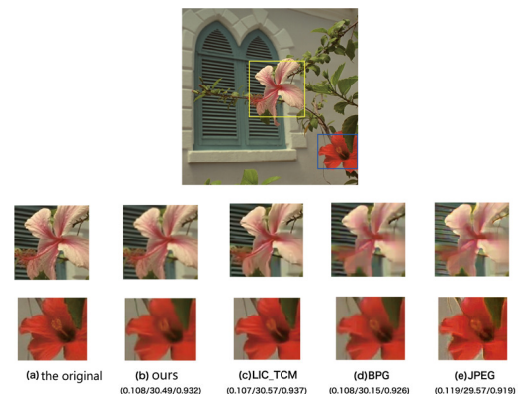
In addition to objective metrics, it is essential to assess subjective visual quality. We perform a visual comparison by compressing two representative images from the Kodak dataset, kodim24 and kodim07, with the results shown in Figures 16 and 17.

In Figure 16, our method delivers markedly better visual quality than JPEG, which suffers from severe blocking artifacts and detail loss. This artifact reduction is especially

critical for preserving the readability of text overlays in digital images. Compared to BPG, our approach yields sharper edges and more consistent colors. Relative to the SOTA method LIC_TCM, our perceptual quality is comparable, yet our method operates at a lower bitrate, demonstrating superior coding efficiency.

**FIGURE 16. Visual comparison of kodim24 by different image compression methods (bpp/PSNR/MS-SSIM).**

Figure 17 further highlights these differences on fine textures. Compared to BPG, both our method and LIC_TCM reconstruct sharper details in the flower petals. For the pink flower, our approach preserves textures that are closer to the ground truth. While the leaf-petal transition in the red flower region is slightly more blurred than in LIC_TCM's reconstruction, this is likely a minor trade-off resulting from our model's emphasis on efficient global context modeling. Our visual analysis focuses on representative images from the Kodak benchmark. Future work could extend this analysis to more specific domains or content types, such as evaluating the preservation of textual regions or facial features on specialized datasets.

**FIGURE 17. Visual comparison of kodim07 by different image compression methods (bpp/PSNR/MS-SSIM).**

E. DISCUSSION AND LIMITATIONS

Our experimental evaluation prioritizes objective metrics of efficiency (e.g., parameter count, inference speed) and

standard distortion measures (e.g., PSNR, MS-SSIM), a focus deliberately aligned with our primary research goal of enabling efficient compression on resource-constrained mobile devices. We acknowledge, however, that these metrics may not fully capture perceptual quality across all content types. While perceptual metrics like LPIPS are a valuable future research direction, this work focuses on PSNR and MS-SSIM to maintain direct and fair comparability with the mainstream compression literature. The current work does not include a perceptual user study, primarily due to the logistical challenges and the lack of a single, universally accepted evaluation protocol for subjective quality. Consequently, a valuable direction for future work is to conduct targeted user studies and to develop content-specific fidelity measures that better assess the practical utility of the compressed images for various applications.

V. CONCLUSION

In this paper, we addressed the critical challenge of efficient learned image compression on resource-constrained devices without introducing additional training cost. Rather than redefining the coding pipeline, our work provides a new efficiency-effectiveness trade-off within the existing CNN-Transformer paradigm. We proposed the LAC framework, a deep learning architecture engineered to meet the demanding constraints of this field. The core of our framework is the PAC module for linear-complexity attention, complemented by the CG unit for reducing channel redundancy and the SHM mechanism for sharing projection matrices. Extensive experiments demonstrate that these modules allow LAC to achieve a superior trade-off between rate-distortion performance and complexity, delivering compression efficiency competitive with state-of-the-art methods while requiring 40% fewer parameters and 13.8% less inference time. This positions LAC as a practical solution for computationally-limited mobile and edge devices, marking a clear distinction from high-complexity models designed for server-side processing.

These advantages make LAC a practical and cost-effective solution for a wide range of modern imaging workflows. Future work will focus on bridging the gap to real-world deployment. This includes integrating the LAC framework into mobile applications for on-device compression and into cloud-based pipelines for large-scale, automated image processing. We will also explore its potential for integration with emerging technologies like 5G Mobile Edge Computing (MEC) to enable ultra-low-latency streaming. Furthermore, extending the framework to video compression and addressing practical challenges such as robust watermarking and end-to-end encryption remain important directions for future research.

REFERENCES

- [1] G. K. Wallace, "The JPEG still picture compression standard," *Commun. ACM*, vol. 34, no. 4, pp. 30–44, Apr. 1991.
- [2] D. Taubman and M. W. Marcellin, "JPEG2000 image compression fundamentals, standards and practice," *J. Electron. Imag.*, vol. 11, no. 2, pp. 286–287, 2002.
- [3] B. Bross, Y.-K. Wang, Y. Ye, S. Liu, J. Chen, G. J. Sullivan, and J.-R. Ohm, "Overview of the versatile video coding (VVC) standard and its applications," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 10, pp. 3736–3764, Oct. 2021.
- [4] Z. Cheng, H. Sun, M. Takeuchi, and J. Katto, "Learned image compression with discretized Gaussian mixture likelihoods and attention modules," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 7936–7945.
- [5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.
- [6] J. Liu, H. Sun, and J. Katto, "Learned image compression with mixed transformer-CNN architectures," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14388–14397.
- [7] L. Zhou, C. Cai, Y. Gao, S. Su, and J. Wu, "Variational autoencoder for low bit-rate image compression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2018, pp. 2617–2620.
- [8] T. Chen, H. Liu, Z. Ma, Q. Shen, X. Cao, and Y. Wang, "End-to-end learnt image compression via non-local attention optimization and improved context modeling," *IEEE Trans. Image Process.*, vol. 30, pp. 3179–3191, 2021.
- [9] W. Li, W. Sun, Y. Zhao, Z. Yuan, and Y. Liu, "Deep image compression with residual learning," *Appl. Sci.*, vol. 10, no. 11, p. 4023, Jun. 2020.
- [10] L. Xiao, H. Wang, and N. Ling, "Image compression with deeper learned transformer," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Nov. 2019, pp. 53–57.
- [11] J. Ballé, D. Minnen, S. Singh, S. J. Hwang, and N. Johnston, "Variational image compression with a scale hyperprior," in *Proc. Int. Conf. Learn. Represent.*, 2018. [Online]. Available: <https://openreview.net/forum?id=rkcQFMZRB>
- [12] G. Zhang, X. Zhang, and L. Tang, "Unified and scalable deep image compression framework for human and machine," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 20, no. 10, pp. 1–22, Oct. 2024, doi: 10.1145/3678472.
- [13] G. Zhang, X. Zhang, and L. Tang, "Enhanced quantified local implicit neural representation for image compression," *IEEE Signal Process. Lett.*, vol. 30, pp. 1742–1746, 2023, doi: 10.1109/LSP.2023.3334956.
- [14] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret, "Transformers are RNNs: Fast autoregressive transformers with linear attention," in *Proc. Int. Conf. Mach. Learn.*, vol. 1, 2020, pp. 5156–5165.
- [15] K. Choromanski, V. Likhoshesterov, D. Dohan, X. Song, A. Gane, T. Sarlos, P. Hawkins, J. Davis, A. Mohiuddin, L. Kaiser, D. Belanger, L. Colwell, and A. Weller, "Rethinking attention with performers," 2020, *arXiv:2009.14794*.
- [16] G. J. Sullivan, J.-R. Ohm, W.-J. Han, and T. Wiegand, "Overview of the high efficiency video coding (HEVC) standard," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 22, no. 12, pp. 1649–1668, Dec. 2012.
- [17] D. Minnen, J. Ballé, and G. D. Toderici, "Joint autoregressive and hierarchical priors for learned image compression," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–8.
- [18] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229.
- [19] Y. Zhu, Y. Yang, and T. Cohen, "Transformer-based transform coding," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–9.
- [20] Z. Qin, W. Sun, H. Deng, D. Li, Y. Wei, B. Lv, J. Yan, L. Kong, and Y. Zhong, "CosFormer: Rethinking softmax in attention," 2022, *arXiv:2202.08791*.
- [21] X. Ma, X. Kong, S. Wang, C. Zhou, J. May, H. Ma, and L. Zettlemoyer, "Luna: Linear unified nested attention," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 2441–2453.
- [22] D. Han, X. Pan, Y. Han, S. Song, and G. Huang, "FLatten transformer: Vision transformer using focused linear attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 5938–5948.
- [23] J. Chen, S.-H. Kao, H. He, W. Zhuo, S. Wen, C.-H. Lee, and S.-H.-G. Chan, "Run, don't walk: Chasing higher FLOPS for faster neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12021–12031.
- [24] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

- [25] M. Lu, F. Chen, S. Pu, and Z. Ma, "High-efficiency lossy image coding through adaptive neighborhood information aggregation," 2022, *arXiv:2204.11448*.
- [26] G. Gao, P. You, R. Pan, S. Han, Y. Zhang, Y. Dai, and H. Lee, "Neural image compression via attentional multi-scale back projection and frequency decomposition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 14657–14666.
- [27] D. Minnen and S. Singh, "Channel-wise autoregressive entropy models for learned image compression," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 3339–3343.
- [28] Z. Guo, Z. Zhang, R. Feng, and Z. Chen, "Causal contextual prediction for learned image compression," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 4, pp. 2329–2341, Apr. 2022.
- [29] H. Liu, T. Chen, P. Guo, Q. Shen, X. Cao, Y. Wang, and Z. Ma, "Non-local attention optimized deep image compression," 2019, *arXiv:1904.09757*.
- [30] C. Ma, Z. Wang, R. Liao, and Y. Ye, "A cross channel context model for latents in deep image compression," 2021, *arXiv:2103.02884*.
- [31] D. He, Y. Zheng, B. Sun, Y. Wang, and H. Qin, "Checkerboard context model for efficient learned image compression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14766–14775.
- [32] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10347–10357.
- [33] J. Liu, G. Lu, Z. Hu, and D. Xu, "A unified end-to-end framework for efficient deep image compression," 2020, *arXiv:2002.03370*.
- [34] G. Toderici, W. Shi, R. Timofte, L. Theis, J. Balle, E. Agustsson, N. Johnston, and F. Mentzer, "Workshop and challenge on learned image compression (clic2020)," in *Proc. CVPR*, Jun. 2020.
- [35] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [36] G. J. Sullivan and T. Wiegand, "Rate-distortion optimization for video compression," *IEEE Signal Process. Mag.*, vol. 15, no. 6, pp. 74–90, Jun. 1998.
- [37] Z. Wang, E. P. Simoncelli, and A. C. Bovik, "Multiscale structural similarity for image quality assessment," in *Proc. 37th Asilomar Conf. Signals, Syst. Comput.*, 2003, pp. 1398–1402.
- [38] G. Bjontegaard, "Calculation of average PSNR differences between RD-curves," ITU-T, Tech. Rep., 2001.
- [39] F. Chen, Y. Xu, and L. Wang, "Two-stage octave residual network for end-to-end image compression," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 4, pp. 3922–3929.

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